

2.10.45

EE25BTECH11016 - Taraka Abhinav

Question:

If $\mathbf{a}$ and $\mathbf{b}$ are two unit vectors such that $\mathbf{a} + 2\mathbf{b}$ and $5\mathbf{a} - 4\mathbf{b}$ are perpendicular to each other then the angle between $\mathbf{a}$ and $\mathbf{b}$ is

a) 45°

b) 60°

c) $\arccos \frac{1}{3}$

d) $\arccos \frac{2}{7}$

Solution:

Let the vectors $\mathbf{a}$ and $\mathbf{b}$ be represented by column matrices. The condition that $\mathbf{a}$ and $\mathbf{b}$ are unit vectors translates to their squared magnitudes being 1. In matrix notation, this is:

$$\mathbf{a}^T \mathbf{a} = 1 \quad \text{and} \quad \mathbf{b}^T \mathbf{b} = 1 \quad (1)$$

The condition that the vectors $(\mathbf{a} + 2\mathbf{b})$ and $(5\mathbf{a} - 4\mathbf{b})$ are perpendicular means their dot product is zero. In matrix notation, the dot product is represented by a matrix product:

$$(\mathbf{a} + 2\mathbf{b})^T (5\mathbf{a} - 4\mathbf{b}) = 0 \quad (2)$$

Expanding the transpose and the product in equation (2):

$$\begin{aligned} (\mathbf{a}^T + 2\mathbf{b}^T)(5\mathbf{a} - 4\mathbf{b}) &= 0 \\ 5(\mathbf{a}^T \mathbf{a}) - 4(\mathbf{a}^T \mathbf{b}) + 10(\mathbf{b}^T \mathbf{a}) - 8(\mathbf{b}^T \mathbf{b}) &= 0 \end{aligned} \quad (3)$$

Since the dot product is commutative, $\mathbf{a}^T \mathbf{b} = \mathbf{b}^T \mathbf{a}$. We can simplify equation (3):

$$5(\mathbf{a}^T \mathbf{a}) + 6(\mathbf{a}^T \mathbf{b}) - 8(\mathbf{b}^T \mathbf{b}) = 0 \quad (4)$$

Now, we substitute the unit vector conditions from equation (1) into equation (4):

$$\begin{aligned} 5(1) + 6(\mathbf{a}^T \mathbf{b}) - 8(1) &= 0 \\ 6(\mathbf{a}^T \mathbf{b}) - 3 &= 0 \\ \mathbf{a}^T \mathbf{b} &= \frac{1}{2} \end{aligned} \quad (5)$$

The angle θ between the vectors is defined by the relation $\mathbf{a}^T \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos \theta$. Using the dot product value from (5) and the magnitudes from (1):

$$\cos \theta = \frac{\mathbf{a}^T \mathbf{b}}{|\mathbf{a}||\mathbf{b}|} = \frac{1/2}{(1)(1)} = \frac{1}{2} \quad (6)$$

Therefore, the angle is:

$$\theta = 60^\circ$$

The correct option is (b).

